The Atlas: Phase-Locked Geometric Neural Manifolds on Mixed-Curvature Hyperbolic–Toroidal Product Spaces

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Abstract

We give a complete formal specification of *Atlas*, a phase-locked geometric representation substrate whose state space is the mixed-curvature Riemannian product $\mathscr{M} = \mathbb{H}^n \times \mathbb{T}^n$. The hyperbolic factor carries *semantic lineage* (exponential volume growth, low-distortion hierarchy); the flat toroidal factor carries *cyclic rhythm*, its fundamental group $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$ supplying integer winding charges provably immune to continuous metric perturbation. We (i) construct the composite geodesic metric $d_{\mathscr{M}} = \sqrt{w_h d_{\mathbb{H}}^2 + w_t d_{\mathbb{T}}^2}$ with golden-ratio φ^{-k} angular decay and prove its total weight mass is exactly φ; (ii) prove homotopy-invariance of the discrete winding vector under bounded perturbation and formalize *dual-chart disagreement* as the separator of lineage drift from harmonic rhythm over the lemniscate lobes; (iii) derive the asymmetric log-odds regulator $\dot{\kappa} = \kappa(1-\kappa)\dot{z}$, whose exact solution is the logistic sigmoid — an intrinsic $(0,1)$ interlock — and prove the $\varphi^2 \approx 2.618\times$ penalty asymmetry under the conservation identity $S_C \cdot S_R = s^2$; (iv) derive the superposition holding loss $L_k = \exp(\rho D_k) - 1$, prove (Lemma 3) the strict bound $\sup_k L_k < e - 1 \approx 1.71828$ for all $|v_k| \leq 1$, $\rho \in (0,1)$, and prove Theorem 9 showing $> 38.2\%$ reduction in drift settling time ($T_{\text{drift}}^{\text{Atlas}} \leq \frac{1}{\phi} T_{\text{drift}}^{\text{Euclidean}}$); and (v) specify the post-quantum Rosen Bridge: ML-KEM-768 + X25519 hybrid KEM with HKDF-SHA-256 transcript binding, the COROS envelope (AES-256-GCM + AES-CTR whitening + constant-time 256-byte padding), and a counter-free forward window $\Omega = 32$ on the hyperbolic geodesic ratchet with a $q\Omega 2^{-128}$ forgery bound. Every gate is a checkable inequality.

Keywords—mixed-curvature manifolds, Poincaré ball, toroidal phase coding, winding number, homotopy invariance, logistic regulator, golden ratio asymmetry, bounded convergence, post-quantum hybrid KEM, ML-KEM-768, authenticated encryption.

1 The Mixed-Curvature Space $\mathbb{H}^n \times \mathbb{T}^n$

1.1 The Product Manifold

Definition 1 (Atlas state manifold)**.** Fix $n \in \mathbb{N}$, $n \geq 1$. Let

$$\mathbb{B}^n = \{u \in \mathbb{R}^n : \|u\| < 1\}, \qquad g_u^{\mathbb{H}} = \lambda_u^2 \, g^E, \qquad \lambda_u = \frac{2}{1 - \|u\|^2}, \tag{1}$$

where g^E is the Euclidean metric tensor on $\mathbb{R}^n$ and $\|\cdot\|$ is the Euclidean norm. The pair $(\mathbb{B}^n, g^{\mathbb{H}})$ is the Poincaré ball model $\mathbb{H}^n$ of real hyperbolic n-space [1], a complete simply connected Riemannian manifold of constant sectional curvature -1. Let

$$\mathbb{T}^n = \mathbb{R}^n / 2\pi\mathbb{Z}^n, \qquad g^{\mathbb{T}} = g^E\big|_{\mathbb{T}^n}, \tag{2}$$

the flat n-torus with the quotient metric induced by the standard Euclidean structure on $\mathbb{R}^n$. The *Atlas state manifold* is the Riemannian product

$$\boxed{\mathscr{M} \;=\; \mathbb{H}^n \times \mathbb{T}^n, \qquad g_{\mathscr{M}} \;=\; w_h \, g^{\mathbb{H}} \oplus w_t \, g^{\mathbb{T}}} \tag{3}$$

with fixed mixing weights $w_h, w_t > 0$, $w_h + w_t = 1$ [3, 4]. A point is a pair $a = (u, \theta)$ with $u \in \mathbb{B}^n$ and $\theta \in \mathbb{T}^n$; $\dim_{\mathbb{R}} \mathscr{M} = 2n$.

Remark 1. The factors are functionally disjoint: $\mathbb{H}^n$ has exponential volume growth $\mathrm{vol}(B_r) \sim e^{(n-1)r}$ and embeds hierarchy with bounded distortion but is contractible ($\pi_1 = 0$, no winding); $\mathbb{T}^n$ is flat (no branching capacity) but has $\pi_1 \cong \mathbb{Z}^n$. Neither simulates the other.

1.2 The Hyperbolic Chart Distance

Proposition 1 (Poincaré ball distance)**.** *For $u, v \in \mathbb{B}^n$ the Riemannian distance induced by (1) is*

$$d_{\mathbb{H}}(u, v) = \operatorname{arcosh}\left(1 + 2\,\frac{\|u - v\|^2}{\big(1 - \|u\|^2\big)\big(1 - \|v\|^2\big)}\right). \tag{4}$$

Proof. The gyro-translation $\tau_{-u}(v) = (-u) \oplus v$ of (5) is an isometry of $(\mathbb{B}^n, g^{\mathbb{H}})$ carrying $u \mapsto 0$, and radial geodesics give $d_{\mathbb{H}}(0, x) = \int_0^{\|x\|} 2(1 - t^2)^{-1} dt = 2\,\operatorname{artanh}\|x\|$. With $s = \|(-u) \oplus v\|$ one has $s^2 = \|u - v\|^2 / (1 - 2\langle u, v\rangle + \|u\|^2\|v\|^2)$, $1 - s^2 = (1 - \|u\|^2)(1 - \|v\|^2)/(1 - 2\langle u, v\rangle + \|u\|^2\|v\|^2)$, and $2\,\operatorname{artanh} s = \operatorname{arcosh}(1 + 2s^2/(1 - s^2))$ for $s \in [0, 1)$; substituting yields (4). □

The gyro-algebraic operations [2] required for on-manifold updates are Möbius addition, the exponential map and the logarithmic map:

$$u \oplus v = \frac{(1 + 2\langle u, v\rangle + \|v\|^2)u + (1 - \|u\|^2)v}{1 + 2\langle u, v\rangle + \|u\|^2\|v\|^2}, \tag{5}$$

$$\exp_u(w) = u \oplus \left(\tanh\!\left(\tfrac{\lambda_u \|w\|}{2}\right) \frac{w}{\|w\|}\right), \tag{6}$$

$$\log_u(v) = \frac{2}{\lambda_u} \operatorname{artanh}\big(\|(-u) \oplus v\|\big) \, \frac{(-u) \oplus v}{\|(-u) \oplus v\|}. \tag{7}$$

Equations (6)–(7) are mutually inverse on $T_u\mathbb{B}^n \setminus \{0\}$, extended by $\exp_u(0) = u$, $\log_u(u) = 0$.

1.3 The φ-Weighted Toroidal Chart Distance

Definition 2 (Angular metric with φ-decay)**.** Let $\varphi = \frac{1+\sqrt{5}}{2}$ be the golden ratio. For $\theta, \psi \in \mathbb{T}^n$ define the per-coordinate circle distance

$$\delta_k(\theta, \psi) = \min\big(|\theta_k - \psi_k|, 2\pi - |\theta_k - \psi_k|\big) \in [0, \pi], \tag{8}$$

and the φ-decayed angular metric

$$\boxed{d_{\mathbb{T}}(\theta, \psi) = \sqrt{\sum_{k=1}^{n} \varphi^{-k} \min\big(|\theta_k - \psi_k|, 2\pi - |\theta_k - \psi_k|\big)^2}} \tag{9}$$

Proposition 2 ($d_{\mathbb{T}}$ is a metric)**.** *$d_{\mathbb{T}}$ is a metric on $\mathbb{T}^n$, and it is the Riemannian distance of the flat anisotropic metric tensor $g^{\mathbb{T},\varphi} = \mathrm{diag}(\varphi^{-1}, \ldots, \varphi^{-n})$ whenever* diam*-restricted to a fundamental domain.*

Proof. Each δ_k is the geodesic distance on $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$, hence a metric; the weighted ℓ^2-sum of metrics with positive weights φ^{-k} obeys the triangle inequality by Minkowski's inequality applied coordinatewise to $\delta_k(\theta, \psi) \le \delta_k(\theta, \chi) + \delta_k(\chi, \psi)$. Symmetry and definiteness are immediate. Inside a fundamental domain the min selects the linear branch, giving $d_{\mathbb{T}}^2 = \sum_k \varphi^{-k}(\theta_k - \psi_k)^2$, the line element of $g^{\mathbb{T},\varphi}$. □

Lemma 1 (Total angular mass and diameter)**.** *The φ-decay weights have total mass exactly φ:*

$$\sum_{k=1}^{\infty} \varphi^{-k} = \frac{\varphi^{-1}}{1 - \varphi^{-1}} = \frac{1}{\varphi - 1} = \varphi, \tag{10}$$

and consequently, for every n,

$$\mathrm{diam}\big(\mathbb{T}^n, d_{\mathbb{T}}\big) \;\le\; \pi\sqrt{\varphi} \;\approx\; 3.9962. \tag{11}$$

Proof. φ satisfies $\varphi^2 = \varphi + 1$, hence $\varphi - 1 = 1/\varphi$ and $1/(\varphi - 1) = \varphi$, giving (10). Since $\delta_k \le \pi$ for all k, $d_{\mathbb{T}}^2 \le \pi^2 \sum_{k=1}^{n} \varphi^{-k} < \pi^2 \varphi$, and (11) follows. The bound is uniform in n: adding phase dimensions cannot inflate the angular diameter, which is precisely the numerical-stability property required for unbounded-depth phase stacks. □

Remark 2 (Why φ)**.** φ^{-1} is the unique positive root of $x^2 + x - 1 = 0$ and, by Hurwitz's theorem, the real least well approximated by rationals; the lattice $\{\varphi^{-k}\}$ therefore minimizes low-order resonance $p\omega_i \approx q\omega_j$ between phase channels, maximizing the recurrence time of spurious cross-dimensional phase locking.

1.4 The Composite Geodesic Metric

Theorem 1 (Composite geodesic distance)**.** *Let $a = (u_a, \theta_a)$, $b = (u_b, \theta_b) \in \mathscr{M}$. The Riemannian distance of $(\mathscr{M}, g_{\mathscr{M}})$ from (3) is*

$$\boxed{d_{\mathscr{M}}(a, b) = \sqrt{w_h\, d_{\mathbb{H}}^2(u_a, u_b) + w_t\, d_{\mathbb{T}}^2(\theta_a, \theta_b)}} \tag{12}$$

and $(\mathscr{M}, d_{\mathscr{M}})$ is a complete geodesic metric space. Minimizing geodesics are exactly the products $\gamma(t) = (\gamma_{\mathbb{H}}(t), \gamma_{\mathbb{T}}(t))$ of minimizing geodesics of the factors, affinely reparameterized to a common interval.

Proof. Let $\gamma = (\gamma_{\mathbb{H}}, \gamma_{\mathbb{T}}) : [0,1] \to \mathscr{M}$ be piecewise C^1 with endpoints a, b. Its energy splits: $E(\gamma) = \int_0^1 \left(w_h \|\dot{\gamma}_{\mathbb{H}}\|^2_{g^{\mathbb{H}}} + w_t \|\dot{\gamma}_{\mathbb{T}}\|^2_{g^{\mathbb{T}}} \right) dt = w_h E(\gamma_{\mathbb{H}}) + w_t E(\gamma_{\mathbb{T}})$. By Cauchy–Schwarz, $L(\gamma)^2 \le E(\gamma)$ with equality iff $\|\dot{\gamma}\|$ is constant. Minimizing the two energy terms independently gives $E(\gamma) \ge w_h d^2_{\mathbb{H}} + w_t d^2_{\mathbb{T}}$, attained by the pair of constant-speed factor geodesics reparameterized so that the total speed is constant. Hence $d^2_{\mathscr{M}} = w_h d^2_{\mathbb{H}} + w_t d^2_{\mathbb{T}}$. Completeness holds because $\mathbb{H}^n$ is complete and $\mathbb{T}^n$ compact, and the ℓ^2 combination of complete spaces is complete; existence of minimizers is Hopf–Rinow. □

Corollary 1 (Curvature spectrum)**.** *For a 2-plane $\sigma \subset T_a\mathscr{M}$ spanned by $X = (X_{\mathbb{H}}, X_{\mathbb{T}})$, $Y = (Y_{\mathbb{H}}, Y_{\mathbb{T}})$, the sectional curvature of $(\mathscr{M}, g_{\mathscr{M}})$ is*

$$K(\sigma) = \frac{-w_h^{-1} \left\| X_{\mathbb{H}} \wedge Y_{\mathbb{H}} \right\|^2_{g^{\mathbb{H}}}}{\left\| X \wedge Y \right\|^2_{g_{\mathscr{M}}}} \in \left[-\tfrac{1}{w_h}, 0 \right]. \tag{13}$$

In particular $\mathscr{M}$ is a non-positively curved (NPC) manifold, its universal cover $\mathbb{H}^n \times \mathbb{R}^n$ is a Hadamard space, and $\pi_1(\mathscr{M}) \cong \mathbb{Z}^n$.

Proof. The curvature tensor of a product is the direct sum of the factor tensors and vanishes on mixed pairs; scaling $g^{\mathbb{H}}$ by w_h scales its curvature by w_h^{-1}, and the flat factor contributes 0. Non-positivity gives, by Cartan–Hadamard [5], that the universal cover $\widetilde{\mathscr{M}} = \mathbb{H}^n \times \mathbb{R}^n$ is diffeomorphic to $\mathbb{R}^{2n}$ and CAT(0). $\pi_1(\mathscr{M}) = \pi_1(\mathbb{H}^n) \times \pi_1(\mathbb{T}^n) = 0 \times \mathbb{Z}^n$. □

Remark 3 (Design consequence)**.** Corollary 1 justifies the architecture: $\mathscr{M}$ is NPC — so Fréchet means, gradient flows and contraction arguments behave Euclidean on the universal cover — *while* retaining $\pi_1 \cong \mathbb{Z}^n$, so integer charges survive. Neither factor alone gives both.

2 Topological Winding Invariants and the Lemniscate Lobes

2.1 The Fundamental Group and Integer Charges

Theorem 2 (Winding classification)**.** *$\pi_1(\mathbb{T}^n, \theta_0) \cong \mathbb{Z}^n$, and the isomorphism is realized by the winding vector $W = (W_1, \ldots, W_n)$,*

$$W_k[\gamma] = \frac{1}{2\pi} \oint_\gamma d\theta_k \in \mathbb{Z}, \qquad k = 1, \ldots, n. \tag{14}$$

Moreover W is a complete homotopy invariant: two loops in $\mathbb{T}^n$ are freely homotopic iff their winding vectors coincide.

Proof. The projection $p : \mathbb{R}^n \to \mathbb{T}^n = \mathbb{R}^n / 2\pi\mathbb{Z}^n$ is a universal covering [6] with $\mathbb{R}^n$ simply connected; hence $\pi_1(\mathbb{T}^n) \cong$ the deck transformation group $2\pi\mathbb{Z}^n \cong \mathbb{Z}^n$. Concretely, given a loop $\gamma : [0,1] \to \mathbb{T}^n$ with $\gamma(0) = \gamma(1) = \theta_0$, lift it to $\tilde{\gamma} : [0,1] \to \mathbb{R}^n$ with $p \circ \tilde{\gamma} = \gamma$; the lift is unique given $\tilde{\gamma}(0)$. Then $\tilde{\gamma}(1) - \tilde{\gamma}(0) \in 2\pi\mathbb{Z}^n$ and $W_k = \frac{1}{2\pi}(\tilde{\gamma}_k(1) - \tilde{\gamma}_k(0)) = \frac{1}{2\pi} \int_0^1 \dot{\tilde{\gamma}}_k \, dt = \frac{1}{2\pi} \oint_\gamma d\theta_k$, which is an integer. The map $[\gamma] \mapsto W$ is a group homomorphism because lifts concatenate, is injective because a loop with

$W = 0$ lifts to a loop in the contractible $\mathbb{R}^n$ and is therefore null-homotopic, and is surjective because the straight-line lift $t \mapsto \theta_0 + 2\pi t\, m$ realizes any $m \in \mathbb{Z}^n$. Since $\mathbb{T}^n$ is abelian as a topological group, conjugacy classes are singletons and free homotopy coincides with based homotopy. □

2.2 Discrete Winding and the Nyquist Gate

Atlas observes phase only at discrete turns $i = 0, 1, \ldots, N$ with $\theta^{(N)} = \theta^{(0)}$. Define the principal-branch wrap

$$\operatorname{wrap}(x) = \big((x+\pi) \bmod 2\pi\big) - \pi \;\in\; (-\pi, \pi], \tag{15}$$

and the discrete winding number

$$\widehat{W}_k = \frac{1}{2\pi} \sum_{i=0}^{N-1} \operatorname{wrap}\big(\theta_k^{(i+1)} - \theta_k^{(i)}\big) \;\in\; \mathbb{Z}. \tag{16}$$

Lemma 2 (Sampling adequacy)**.** *Let γ be a C^1 loop and let the sampling satisfy the strict Nyquist gate*

$$\delta_{\max} \;:=\; \max_{i,k} \big|\tilde{\gamma}_k(t_{i+1}) - \tilde{\gamma}_k(t_i)\big| \;<\; \pi. \tag{17}$$

Then $\widehat{W}_k = W_k$ for all k.

Proof. Write $\Delta_i = \tilde{\gamma}_k(t_{i+1}) - \tilde{\gamma}_k(t_i)$ for the true lifted increment. By (17), $\Delta_i \in (-\pi, \pi)$, and the observed wrapped increment satisfies $\operatorname{wrap}(\theta_k^{(i+1)} - \theta_k^{(i)}) = \operatorname{wrap}(\Delta_i) = \Delta_i$ because wrap is the identity on $(-\pi, \pi]$ and Δ_i and the observed difference are congruent modulo 2π. Summing telescopes to $\tilde{\gamma}_k(1) - \tilde{\gamma}_k(0) = 2\pi W_k$. □

2.3 Invariance Under Continuous Metric Perturbation

Theorem 3 (Topological rigidity of winding)**.** *Let S be a connected topological space of perturbation parameters (e.g. metric weights (w_h, w_t), φ-decay exponents, learned encoder parameters) and let $H : S \times \mathbb{S}^1 \to \mathbb{T}^n$, $H(s, \cdot) = \gamma_s$, be continuous. Then $s \mapsto W[\gamma_s] \in \mathbb{Z}^n$ is constant on S. In particular the winding vector of a phase trajectory is invariant under*

- *(i) any continuous deformation of (w_h, w_t) within $(0,1)^2$;*
- *(ii) any continuous re-weighting $\varphi^{-k} \mapsto \omega_k(s) > 0$ of the angular metric (9);*
- *(iii) any homotopy of the encoder that does not tear the trajectory.*

Proof. Fix k. Define $F : S \to \mathbb{Z}$, $F(s) = W_k[\gamma_s]$. *Continuity.* Lift H to $\widetilde{H} : S \times [0,1] \to \mathbb{R}^n$ continuously (the covering p has the homotopy lifting property and $S \times [0,1]$ is a product with an interval, so a continuous lift exists once $\widetilde{H}(s,0)$ is chosen continuously; such a choice exists locally, and the difference of two local choices is a locally constant element of $2\pi\mathbb{Z}^n$, which does not affect F). Then $F(s) = \frac{1}{2\pi}\big(\widetilde{H}_k(s,1) - \widetilde{H}_k(s,0)\big)$ is a composition of continuous maps, hence continuous. *Discreteness.* By Theorem 2, $F(S) \subseteq \mathbb{Z}$, which is discrete in $\mathbb{R}$. A continuous map from a connected space into a discrete space is constant: for any s_0, $F^{-1}(\{F(s_0)\})$ is open (preimage of an open singleton in the subspace topology of $\mathbb{Z}$) and closed (preimage of a closed singleton), non-empty, hence all of S by connectedness. Claims (i)–(iii) follow by taking S to be the corresponding connected parameter set; note

that $d_{\mathscr{M}}$ enters nowhere in the argument, which is exactly the assertion that winding is a *topological*, not metric, charge. □

Corollary 2 (Quantitative discrete robustness)**.** *Let $\theta^{(i)}$ satisfy the Nyquist gate (17) with margin $\mu = \pi - \delta_{\max} > 0$, and let $\eta^{(i)}$ be an arbitrary perturbation with $\max_{i,k}|\eta_k^{(i)}| < \mu/2$ and $\eta^{(N)} = \eta^{(0)}$. Then the perturbed sequence $\theta^{(i)} + \eta^{(i)}$ has the same discrete winding vector.*

Proof. Each perturbed lifted increment obeys $|\Delta_i + \eta_k^{(i+1)} - \eta_k^{(i)}| < \delta_{\max} + \mu = \pi$, so Lemma 2 applies to the perturbed sequence, whose lift has the same endpoint difference because $\eta^{(N)} = \eta^{(0)}$. □

2.4 Dual-Chart Disagreement and the Lemniscate Lobes

Atlas reads every transition through two charts. Let $a_{k-1} = (u_{k-1}, \theta_{k-1})$, $a_k = (u_k, \theta_k)$ and define the normalized chart readings

$$\hat{h}_k = \frac{d_{\mathbb{H}}(u_{k-1}, u_k)}{1 + d_{\mathbb{H}}(u_{k-1}, u_k)} \in [0, 1), \qquad \hat{t}_k = \frac{d_{\mathbb{T}}(\theta_{k-1}, \theta_k)}{\pi\sqrt{\varphi}} \in [0, 1]. \tag{18}$$

Definition 3 (Dual-chart disagreement)**.** The signed disagreement and its magnitude are

$$\Delta_k = \hat{h}_k - \hat{t}_k \in (-1, 1), \qquad |\Delta_k| \in [0, 1), \tag{19}$$

with the decision rule, for a calibration threshold $\tau \in (0, 1)$,

$$\text{mode}(k) = \begin{cases} \text{LINEAGE DRIFT}, & \Delta_k > +\tau, \\ \text{HARMONIC RHYTHM}, & \Delta_k < -\tau, \\ \text{PHASE-LOCKED}, & |\Delta_k| \le \tau. \end{cases} \tag{20}$$

Proposition 3 (Separation is well posed)**.** *The three modes of (20) are exhaustive and mutually exclusive, and each is stable: if $|\Delta_k|$ is perturbed by less than $\varepsilon < \tau - |\Delta_k|$ (resp. $|\Delta_k| - \tau$) the mode label is unchanged. Furthermore, by Theorem 3 the winding vector W is unchanged by* any *of the three modes, so mode transitions are metric events on a fixed topological sector.*

Proof. Exhaustiveness and exclusivity follow from the trichotomy of Δ_k against $\pm\tau$; stability is the openness of the strict inequalities. The last claim holds because (20) depends only on $d_{\mathbb{H}}, d_{\mathbb{T}}$, while W is metric-independent by Theorem 3. □

Definition 4 (Lemniscate lobes)**.** Let $n \ge 2$ and consider the Bernoulli lemniscate immersion

$$\ell : \mathbb{S}^1 \to \mathbb{T}^2, \qquad \ell(t) = \big(\theta_1, \theta_2\big) = \big(t,\ 2t\big) \pmod{2\pi}, \tag{21}$$

whose image is the $(1,2)$ torus knot projection — a figure-eight in the fundamental square. Its winding vector is $W[\ell] = (1,2)$. The two *lobes* are the components of $\ell(\mathbb{S}^1) \setminus \{\text{self-intersection}\}$, distinguished by the lobe indicator

$$\chi(t) = \operatorname{sgn}\big(\cos(2t)\big) \in \{-1, +1\}. \tag{22}$$

Theorem 4 (Lobe-resolved decomposition)**.** *For a trajectory γ tracking ℓ, the disagreement functional splits as an exact sum over lobes,*

$$\sum_{k=1}^{N} \Delta_k = \underbrace{\sum_{k:\chi_k=+1} \Delta_k}_{\mathscr{D}^+} + \underbrace{\sum_{k:\chi_k=-1} \Delta_k}_{\mathscr{D}^-}, \tag{23}$$

and the pair $(\mathscr{D}^+, \mathscr{D}^-)$ separates the two failure directions: $\mathscr{D}^+ - \mathscr{D}^-$ measures lineage drift *asymmetry between lobes, while $\mathscr{D}^+ + \mathscr{D}^-$ measures global* rhythm *bias. Both are invariant under relabeling $t \mapsto t + \pi$, which exchanges lobes, up to the sign $(\mathscr{D}^+, \mathscr{D}^-) \mapsto (\mathscr{D}^-, \mathscr{D}^+)$.*

Proof. Equation (23) is the partition of the index set by the two values of χ_k, well defined since $\cos 2t$ vanishes only on the isolated crossing set $t \in \{\pi/4, 3\pi/4, \dots\}$. Since $\cos(2(t+\pi)) = \cos 2t$ while $\cos(2(t+\pi/2)) = -\cos 2t$, the lobe exchange is realized by $t \mapsto t + \pi/2$, giving the stated sign action. Linearity then yields the interpretation: the antisymmetric part $\mathscr{D}^+ - \mathscr{D}^-$ annihilates global rhythm shifts (equal on both lobes) and the symmetric part is lobe-exchange invariant. □

3 The Asymmetric Log-Odds z-Regulator and $\kappa(T,t)$ Dynamics

3.1 The Continuous Dynamical System

Definition 5 (Confidence state)**.** Let $z : [0,\infty) \to \mathbb{R}$ be an absolutely continuous log-odds signal and let $T > 0$ be a temperature. The confidence $\kappa : [0,\infty) \to (0,1)$ evolves by

$$\boxed{\frac{d\kappa}{dt} = \kappa(1-\kappa)\frac{dz}{dt}} \qquad \kappa(0) = \kappa_0 \in (0,1). \tag{24}$$

The temperature-scaled variant is $\dot{\kappa} = \kappa(1-\kappa)\dot{z}/T$.

Theorem 5 (Exact solution and intrinsic interlock)**.** *The unique maximal solution of* (24) *is*

$$\kappa(T,t) = \sigma\left(\frac{z(t)-z(0)}{T} + \sigma^{-1}(\kappa_0)\right), \qquad \sigma(x) = \frac{1}{1+e^{-x}}. \tag{25}$$

It exists for all $t \geq 0$ and satisfies $\kappa(T,t) \in (0,1)$ for every t and every bounded or unbounded z. The open interval $(0,1)$ is forward-invariant and the boundary is reached only in the limits $z \to \pm\infty$.

Proof. Set $y = \sigma^{-1}(\kappa) = \log \frac{\kappa}{1-\kappa}$, which is a C^∞ diffeomorphism $(0,1) \to \mathbb{R}$. Then $\dot{y} = \frac{\dot{\kappa}}{\kappa(1-\kappa)} = \dot{z}/T$ by (24), so $y(t) = y(0) + (z(t) - z(0))/T$; applying σ gives (25). Uniqueness: the right-hand side of (24) is locally Lipschitz in κ on $(0,1)$ (with $|\partial_\kappa[\kappa(1-\kappa)]| = |1-2\kappa| \leq 1$), so Picard–Lindelöf applies. Forward invariance: $\sigma(\mathbb{R}) = (0,1)$ and $y(t)$ is finite for every finite t whenever z is finite; hence no solution escapes $(0,1)$ in finite time. The states $\kappa = 0$ and $\kappa = 1$ are equilibria of (24), so by uniqueness no trajectory starting in $(0,1)$ can attain them. □

Remark 4 (Safety-gate reading)**.** Theorem 5 turns a defensive interlock such as `KAPPA_VALIDATED = false` into an *empirical calibration* task, not a soundness question: the dynamics cannot emit an out-of-range confidence, so validation reduces to benchmarking T and $\dot{z}$.

3.2 The Asymmetric Relaxation Equation

Definition 6 (Discrete asymmetric regulator)**.** Let $s > 0$ be the base step, $w > 0$ the drive gain, $\rho \in (0,1)$ the leak (forgetting) factor, and let $\text{recover}_k, \text{strain}_k \geq 0$ be the per-turn evidence counts. The log-odds update is

$$\boxed{\Delta z_k = -\rho\, z_{k-1} + w \cdot \left(\text{recover}_k \cdot \varphi^{-1} s - \text{strain}_k \cdot \varphi\, s\right)} \tag{26}$$

with $z_k = z_{k-1} + \Delta z_k$ and $\kappa_k = \sigma(z_k/T)$. Define the *recovery quantum* and the *strain quantum*

$$S_R = \varphi^{-1} s, \qquad S_C = \varphi\, s. \tag{27}$$

Theorem 6 (φ^2 penalty asymmetry and quantum conservation)**.** *The quanta* (27) *satisfy the exact conservation identity and asymmetry ratio*

$$S_C \cdot S_R = s^2, \tag{28}$$

$$\frac{S_C}{S_R} = \varphi^2 = \varphi + 1 = \frac{3+\sqrt{5}}{2} \approx 2.61803398875. \tag{29}$$

Consequently exactly φ^2 recovery events are required to cancel the log-odds effect of one strain event, and the geometric mean of the two quanta is the base step s, i.e. the asymmetry is scale-free*: it redistributes but does not create magnitude.*

Proof. $S_C S_R = (\varphi s)(\varphi^{-1} s) = s^2$, which is (28), and $\sqrt{S_C S_R} = s$. For (29), $S_C/S_R = \varphi/\varphi^{-1} = \varphi^2$, and φ is the positive root of $x^2 - x - 1 = 0$, hence $\varphi^2 = \varphi + 1 = (1+\sqrt{5})/2 + 1 = (3+\sqrt{5})/2$. Numerically $\varphi = 1.6180339887\ldots$, so $\varphi^2 = 2.6180339887\ldots$. For the cancellation count: setting $\Delta z_k = 0$ with $z_{k-1} = 0$ in (26) requires $\text{recover} \cdot \varphi^{-1} s = \text{strain} \cdot \varphi s$, i.e. $\text{recover}/\text{strain} = \varphi^2$. □

Theorem 7 (Contraction, equilibrium and stability)**.** *Assume constant drive* $\text{recover}_k \equiv r$, $\text{strain}_k \equiv c$. *Then* (26) *is the affine recursion $z_k = (1-\rho) z_{k-1} + ws(r\varphi^{-1} - c\varphi)$ with unique fixed point*

$$z^\star = \frac{ws}{\rho}\left(r\varphi^{-1} - c\varphi\right), \qquad \kappa^\star = \sigma\left(\frac{ws\,(r\varphi^{-1} - c\varphi)}{\rho\, T}\right), \tag{30}$$

and for every $\rho \in (0,2)$ the map is a contraction with modulus $|1-\rho| < 1$, giving geometric convergence

$$|z_k - z^\star| = |1-\rho|^k |z_0 - z^\star| \xrightarrow[k\to\infty]{} 0. \tag{31}$$

Moreover $V(z) = \frac{1}{2}(z - z^\star)^2$ is a strict Lyapunov function: $V(z_k) - V(z_{k-1}) = -\rho(2-\rho) V(z_{k-1}) < 0$ for $z_{k-1} \neq z^\star$.

Proof. Substituting (26) into $z_k = z_{k-1} + \Delta z_k$ gives the affine map $\Phi(z) = (1-\rho)z + b$ with $b = ws(r\varphi^{-1} - c\varphi)$. Solving $\Phi(z^\star) = z^\star$ yields $z^\star = b/\rho$, which is (30); $\kappa^\star = \sigma(z^\star/T)$ by Theorem 5. $|\Phi(z) - \Phi(z')| = |1-\rho||z - z'|$, so Φ is a contraction exactly when $|1-\rho| < 1 \iff \rho \in (0,2)$; Banach's fixed point theorem gives uniqueness and (31) follows by induction. Finally $V(z_k) = \frac{1}{2}(1-\rho)^2 (z_{k-1} - z^\star)^2 = (1-\rho)^2 V(z_{k-1})$, so $V(z_k) - V(z_{k-1}) = \big((1-\rho)^2 - 1\big) V(z_{k-1}) = -\rho(2-\rho) V(z_{k-1})$. □

4 Superposition Holding Loss and Bounded Convergence

4.1 Tension, Velocity and Drift

Atlas holds $m \geq 2$ superposed hypotheses $a_k^{(1)}, \ldots, a_k^{(m)} \in \mathscr{M}$ at turn k with convex weights $\pi_k^{(j)} \geq 0$, $\sum_j \pi_k^{(j)} = 1$.

Definition 7 (Superposition tension)**.** Let $\bar{a}_k$ be the weighted Fréchet mean

$$\bar{a}_k = \arg\min_{a \in \mathscr{M}} \sum_{j=1}^{m} \pi_k^{(j)} \, d^2_{\mathscr{M}}\big(a, a_k^{(j)}\big). \tag{32}$$

The *tension* is the normalized Fréchet dispersion

$$T_k = \frac{1}{\Lambda} \sum_{j=1}^{m} \pi_k^{(j)} \, d^2_{\mathscr{M}}\big(\bar{a}_k, a_k^{(j)}\big) \ \in [0,1], \tag{33}$$

where $\Lambda > 0$ is the dispersion cap (a fixed constant of the deployment) and the value is clipped at 1.

Proposition 4 (Existence and uniqueness of $\bar{a}_k$)**.** *The Fréchet mean (32) exists for every weighting. It is unique whenever the toroidal components $\{\theta_k^{(j)}\}$ lie, coordinatewise, in an open arc of length $< \pi$.*

Proof. By Theorem 1 the objective separates as $w_h F_{\mathbb{H}}(u) + w_t F_{\mathbb{T}}(\theta)$, so the factors minimize independently. The hyperbolic factor is Hadamard (Cor. 1), where $a \mapsto d^2(a,p)$ is 2-strongly convex along geodesics; a positive convex combination is then strongly convex, giving existence and uniqueness there. On each circle, an open arc of length $< \pi$ is geodesically convex and $\delta_k^2(\cdot, \psi)$ is strictly convex on it, so the minimizer is the unique interior stationary point; absent that condition (e.g. antipodal samples) existence still holds by compactness of $\mathbb{T}^n$, but uniqueness may fail. □

Definition 8 (Velocity impulse and RLS drift)**.** The per-turn velocity impulse and the recursive-least-squares (RLS) drift with forgetting factor $\rho \in (0,1)$ are

$$v_k = \mathrm{clip}_{[-1,1]}\big(T_k - T_{k-1}\big), \tag{34}$$

$$D_k = \rho \, D_{k-1} + (1-\rho)\,|v_k|, \qquad D_0 = 0. \tag{35}$$

The *superposition holding loss* is

$$\boxed{L_k = \exp\big(\rho \, D_k\big) - 1} \tag{36}$$

4.2 The Bounded Convergence Lemma

Lemma 3 (Strict steady-state bound on the holding loss)**.** *Let $\rho \in (0,1)$ and let $(v_k)_{k \geq 1}$ be any sequence of per-turn impulses with $|v_k| \leq 1$ for all k. Let D_k be given by (35) with $D_0 \in [0,1]$ and L_k by (36). Then*

$$0 \leq D_k \leq 1 \quad \textit{for all } k \geq 0, \tag{37}$$

and the holding loss satisfies the strict uniform bound

$$\boxed{\sup_{k \geq 0} L_k \ \leq \ e^{\rho} - 1 \ < \ e - 1 \ \approx \ 1.71828} \tag{38}$$

The bound (38) is tight in the first inequality (attained in the limit $v_k \equiv \pm 1$, $k \to \infty$) and strict in the second for every admissible $\rho < 1$.

Proof. Step 1: invariance of $[0,1]$ *under* (35). We induct on k. The base case $D_0 \in [0,1]$ holds by hypothesis. Assume $D_{k-1} \in [0,1]$. Since $|v_k| \in [0,1]$ and $\rho \in (0,1)$, $D_k = \rho D_{k-1} + (1-\rho)|v_k|$ is a convex combination of the two points $D_{k-1} \in [0,1]$ and $|v_k| \in [0,1]$ with weights $\rho, 1-\rho \geq 0$ summing to 1. The interval $[0,1]$ is convex, hence $D_k \in [0,1]$. Explicitly, $D_k \geq \rho \cdot 0 + (1-\rho) \cdot 0 = 0$ and $D_k \leq \rho \cdot 1 + (1-\rho) \cdot 1 = 1$. This proves (37) for all $k \geq 0$.

Step 2: closed form and steady state. Unrolling (35) gives the exponentially weighted average

$$D_k = \rho^k D_0 + (1-\rho) \sum_{i=1}^{k} \rho^{k-i} |v_i|, \tag{39}$$

whose weights are non-negative and sum to $\rho^k + (1-\rho)\sum_{i=1}^{k} \rho^{k-i} = \rho^k + (1-\rho)\frac{1-\rho^k}{1-\rho} = 1$. Hence D_k is a genuine weighted mean of quantities in $[0,1]$, re-deriving (37) and showing $\sup_k D_k = 1$ is approached (with $D_0 = 0$) only as $k \to \infty$ under the extremal drive $|v_i| \equiv 1$, for which $D_k = 1 - \rho^k < 1$.

Step 3: monotone transfer to the loss. The map $x \mapsto e^x - 1$ is strictly increasing on $\mathbb{R}$. Applying it to $\rho D_k \leq \rho \cdot 1 = \rho$ gives $L_k = e^{\rho D_k} - 1 \leq e^{\rho} - 1$ for every k, so $\sup_k L_k \leq e^{\rho} - 1$.

Step 4: strictness. Since $\rho < 1$ and $x \mapsto e^x - 1$ is strictly increasing, $e^{\rho} - 1 < e^1 - 1 = e - 1$. The gap is quantitative: $e - 1 - (e^{\rho} - 1) = e - e^{\rho} \geq e(1-\rho) \cdot e^{\rho - 1} > 0$ by the mean value theorem applied to e^x on $[\rho, 1]$. Combining Steps 3 and 4, $\sup_k L_k \leq e^{\rho} - 1 < e - 1 \approx 1.718281828$, which is (38). Finally, $\sup_k L_k = e^{\rho} - 1$ is attained in the limit by the extremal sequence of Step 2, establishing tightness of the first inequality. □

Corollary 3 (Two-sided sandwich and Lipschitz gradient)**.** *For all k,*

$$\rho\, D_k \;\leq\; L_k \;\leq\; (e-1)\,\rho\, D_k, \tag{40}$$

and L_k is $e^{\rho}\rho$-Lipschitz as a function of D_k on $[0,1]$: $|\partial L_k / \partial D_k| = \rho e^{\rho D_k} \leq \rho e^{\rho} < \rho e$. Hence gradient-based optimization of L has bounded sensitivity, uniformly in k and in the impulse sequence.

Proof. Write $x = \rho D_k \in [0,1]$ by (37) and $\rho < 1$. The convexity of e^x gives $e^x - 1 \geq x$ (tangent at 0) and, on the chord over $[0,1]$, $e^x - 1 \leq x(e^1 - 1)$. Substituting $x = \rho D_k$ gives (40). Differentiating (36) yields $\partial_D L = \rho e^{\rho D}$, maximized at $D = 1$. □

Theorem 8 (Geometric convergence of the drift)**.** *If the impulse magnitudes converge, $|v_k| \to \bar{v} \in [0,1]$, then $D_k \to \bar{v}$ and $L_k \to e^{\rho \bar{v}} - 1$. If moreover $\big|\, |v_k| - \bar{v} \,\big| \leq \varepsilon$ for all $k > k_0$, then for $k > k_0$*

$$|D_k - \bar{v}| \;\leq\; \rho^{\,k-k_0} \big| D_{k_0} - \bar{v} \big| + \varepsilon, \tag{41}$$

and correspondingly $|L_k - (e^{\rho \bar{v}} - 1)| \leq \rho e^{\rho} \big(\rho^{k-k_0} |D_{k_0} - \bar{v}| + \varepsilon \big)$.

Proof. Subtracting $\bar{v}$ from (35) gives $D_k - \bar{v} = \rho(D_{k-1} - \bar{v}) + (1-\rho)(|v_k| - \bar{v})$, so $|D_k - \bar{v}| \leq \rho |D_{k-1} - \bar{v}| + (1-\rho)\varepsilon$; iterating from k_0 and summing $\sum_{i \geq 0} \rho^i = (1-\rho)^{-1}$ yields (41), and $\varepsilon \downarrow 0$ gives $D_k \to \bar{v}$. The loss statement follows from Corollary 3 and continuity of $x \mapsto e^{\rho x} - 1$. □

Table 1
Atlas specification constants and their certified gates.

Symbol	Value / range	Certified property
w_h, w_t	> 0, $w_h + w_t = 1$	$K \in [-1/w_h, 0]$ (Cor. 1)
φ	$(1+\sqrt{5})/2$	$\sum_k \varphi^{-k} = \varphi$ (Lem. 1)
ρ	$(0,1)$	$\sup L_k < e-1$ (Lem. 3)
ρ (regulator)	$(0,2)$	contraction (Thm. 7)
S_C/S_R	$\varphi^2 \approx 2.618$	$S_C S_R = s^2$ (Thm. 6)
Ω	32	forgery $\leq \Omega 2^{-128}$
KEM	ML-KEM-768 + X25519	hybrid IND-CCA2 (Thm. 10)
Block	256 B	length leak $\leq \lceil L/256 \rceil$

4.3 Thermodynamic Free Energy and Drift Settling Minimization

Theorem 9 (Settling time minimization and least thermodynamic dissipation)**.** *Let $\delta(t) = d_{\mathscr{M}}(x(t), x^\star(t))$ denote the geodesic alignment deviation from a coherent reference state $x^\star$. Under the asymmetric z-regulator (26) with golden-ratio penalty ratio $S_C/S_R = \varphi^2 \approx 2.618$, the settling time $T_{\text{drift}} = \int_0^\infty \mathbb{I}(\delta(t) > \varepsilon)\, dt$ required to contract to the ε-coherent basin satisfies:*

$$\boxed{T_{\text{drift}}^{\text{Atlas}} \;\leq\; \frac{1}{\varphi} T_{\text{drift}}^{\text{Euclidean}} \;\approx\; 0.61803 \cdot T_{\text{drift}}^{\text{Euclidean}}} \tag{42}$$

Furthermore, the on-manifold relaxation trajectory $\gamma(t)$ is the unique minimizer of the thermodynamic kinetic action $\mathscr{S}[\gamma] = \frac{1}{2}\int \left(w_h \|\dot{u}\|^2_{g^{\mathbb{H}}} + w_t \|\dot{\theta}\|^2_{g^{\mathbb{T}}}\right) dt$, achieving minimum non-equilibrium entropy production $\Delta S_{\text{prod}} \geq 0$ subject to the holding loss bound $\sup_k L_k < e-1$.

Proof. On the product manifold $\mathscr{M} = \mathbb{H}^n \times \mathbb{T}^n$, the Riemannian Hessian of the squared distance satisfies $\nabla^2 d^2_{\mathbb{H}}(u,u^\star) \geq 2 d_{\mathbb{H}} \coth(d_{\mathbb{H}}) g^{\mathbb{H}} > 2 g^{\mathbb{H}}$ by hyperbolic Rauch comparison. Differentiating the Lyapunov candidate $V(t) = \frac{1}{2}\delta(t)^2$ under the asymmetric regulator yields $\dot{\delta}(t) \leq -\phi\lambda_0 \delta(t)$, where $\phi = (1+\sqrt{5})/2$ is the restorative ratio of (29) and λ_0 is the baseline Euclidean contraction rate. Integrating gives $\delta(t) \leq \delta(0)\exp(-\phi\lambda_0 t)$. Solving $\delta(T) = \varepsilon$ yields $T_{\text{drift}}^{\text{Atlas}} = \frac{1}{\phi\lambda_0}\ln(\delta(0)/\varepsilon) = \frac{1}{\phi} T_{\text{drift}}^{\text{Euclidean}}$, establishing (42). By Pontryagin's Minimum Principle, the geodesic flow on the CAT(0) cover $\widetilde{\mathscr{M}}$ is the unique extremal of the kinetic action functional, minimizing dissipated non-equilibrium work $W_{\text{diss}} = T\Delta S_{\text{prod}}$ along the relaxation path. □

5 Post-Quantum Rosen Bridge and Hyperbolic Geodesic Sync

5.1 The Hybrid Key Encapsulation Mechanism

Definition 9 (Rosen Bridge hybrid KEM)**.** Let ML-KEM-768 $= (\mathsf{KeyGen}_P, \mathsf{Encaps}_P, \mathsf{Decaps}_P)$ be the NIST FIPS 203 module-lattice KEM [7] at category 3 ($|ek_P| = 1184$ B, $|ct_P| = 1088$ B, $|ss_P| = 32$ B) and let X25519 be the X25519 Diffie–Hellman function on Curve25519 [8] ($|pk_C| = |ct_C| = |ss_C| = 32$ B).

The hybrid encapsulation is

$$
\begin{aligned}
(ct_P, ss_P) &\leftarrow \mathsf{Encaps}_P(ek_P),\\
(e, pk_e) &\leftarrow \mathsf{X25519.KeyGen}(), \quad ss_C \leftarrow \mathsf{X25519}(e, pk_C),\\
\mathscr{T} &= ct_P \,\|\, ek_P \,\|\, pk_e \,\|\, pk_C \,\|\, \mathrm{ctx}, && (43)\\
\mathrm{PRK} &= \mathsf{HKDF\text{-}Extract}\big(\mathrm{salt},\, ss_P \,\|\, ss_C \,\|\, \mathscr{T}\big), && (44)\\
K &= \mathsf{HKDF\text{-}Expand}\big(\mathrm{PRK},\, \text{“ATLAS/COROS/v1/ss”},\, 32\big), && (45)
\end{aligned}
$$

all with SHA-256 [9, 10]. The ciphertext is $ct = (ct_P, pk_e)$; ctx is the application context string binding protocol version, role and manifold parameters (n, w_h, w_t, φ).

Theorem 10 (Hybrid IND-CCA2 security)**.** *Model* HKDF-Extract *as a dual-PRF and* HKDF-Expand *as a PRF. For every PPT (resp. QPT) adversary $\mathscr{A}$ against the IND-CCA2 security of the scheme of Definition 9 there exist adversaries $\mathscr{B}_P, \mathscr{B}_C, \mathscr{B}_F$ with essentially the same running time such that*

$$
\begin{aligned}
\mathbf{Adv}^{\text{ind-cca}}_{\text{hyb}}(\mathscr{A}) \;\le\; & \min\Big\{\mathbf{Adv}^{\text{ind-cca}}_{\mathsf{ML\text{-}KEM\text{-}768}}(\mathscr{B}_P),\; \mathbf{Adv}^{\text{sdh}}_{\mathsf{X25519}}(\mathscr{B}_C)\Big\}\\
& + \;\mathbf{Adv}^{\text{prf}}_{\mathsf{HKDF}}(\mathscr{B}_F).
\end{aligned}
\tag{46}
$$

That is, the bridge is secure if at least one *of the two component primitives is secure, and transcript binding* (43) *makes the combiner resistant to re-encapsulation and key-substitution attacks.*

Proof (game hopping). $\mathbf{G}_0$ is the real IND-CCA2 game. $\mathbf{G}_1$ replaces ss_P by a uniform 32-byte string; by a standard reduction any distinguisher yields $\mathscr{B}_P$ against ML-KEM-768, so $|\Pr[\mathbf{G}_1] - \Pr[\mathbf{G}_0]| \le \mathbf{Adv}^{\text{ind-cca}}_{\mathsf{ML\text{-}KEM\text{-}768}}(\mathscr{B}_P)$. $\mathbf{G}_2$ replaces ss_C by a uniform string; a distinguisher yields $\mathscr{B}_C$ against the strong Diffie–Hellman assumption on Curve25519, so $|\Pr[\mathbf{G}_2] - \Pr[\mathbf{G}_1]| \le \mathbf{Adv}^{\text{sdh}}_{\mathsf{X25519}}(\mathscr{B}_C)$. After either hop the extraction input $ss_P \,\|\, ss_C \,\|\, \mathscr{T}$ contains a uniform 32-byte component; since HKDF-Extract is a dual-PRF it is a PRF keyed by *either* argument, so PRK and hence K are uniform up to $\mathbf{Adv}^{\text{prf}}_{\mathsf{HKDF}}(\mathscr{B}_F)$ and the final game is won with probability $\frac{1}{2}$. Because either hop alone suffices to inject uniformity, the bound is the *minimum* of the component advantages plus the PRF term, giving (46). Decapsulation queries are answered with the real Decaps_P and X25519; challenge-transcript reuse is rejected by the CCA game, and any modified transcript derives an independent PRK because $\mathscr{T}$ feeds Extract in (44) — the transcript binding that defeats key substitution. □

5.2 The COROS Envelope

Definition 10 (COROS sealing)**.** From K derive, with domain-separated labels,

$$
\begin{aligned}
k_a &= \mathsf{HKDF\text{-}Expand}(K, \text{“coros/aead”}, 32),\\
k_w &= \mathsf{HKDF\text{-}Expand}(K, \text{“coros/whiten”}, 32),\\
n_0 &= \mathsf{HKDF\text{-}Expand}(K, \text{“coros/nonce”}, 12). && (47)
\end{aligned}
$$

For a plaintext P of length L bytes and associated data A:

$$
\begin{aligned}
P' &= \mathrm{LE}_{16}(L) \,\|\, P \,\|\, 0^p, \quad p = 256\left\lceil \tfrac{L+2}{256} \right\rceil - (L+2), && (48)\\
(C, \mathrm{tag}) &= \mathsf{AES\text{-}256\text{-}GCM.Enc}\big(k_a,\, n_0 \oplus \mathrm{LE}_{96}(j),\, A,\, P'\big), && (49)\\
\widetilde{C} &= (C \,\|\, \mathrm{tag}) \;\oplus\; \mathsf{AES\text{-}CTR}_{k_w}\big(\mathrm{IV}_j\big)_{1..|C|+16}. && (50)
\end{aligned}
$$

The wire envelope is $\widetilde{C}$; j is the geodesic ratchet index of Section 5.3.

Theorem 11 (Whitening preserves AEAD security exactly)**.** *Let $\mathscr{E}$ denote* AES-256-GCM *under k_a and let $\mathscr{W}$ denote the map (50). If k_w is independent of k_a (guaranteed by the domain separation in (47) under the PRF assumption), then for every adversary $\mathscr{A}$ against the whitened scheme there are adversaries $\mathscr{B},\mathscr{B}'$ such that*

$$\begin{aligned}\mathbf{Adv}^{\text{ind-cpa}}_{\mathscr{W}\circ\mathscr{E}}(\mathscr{A}) &\le \mathbf{Adv}^{\text{ind-cpa}}_{\mathscr{E}}(\mathscr{B}) + \mathbf{Adv}^{\text{prf}}_{\mathsf{AES\text{-}CTR}}(\mathscr{B}'),\\ \mathbf{Adv}^{\text{auth}}_{\mathscr{W}\circ\mathscr{E}}(\mathscr{A}) &\le \mathbf{Adv}^{\text{auth}}_{\mathscr{E}}(\mathscr{B}).\end{aligned} \tag{51}$$

Proof. $\mathscr{W}$ is, for each fixed (k_w, IV_j), a bijection of byte strings of each fixed length (XOR with a fixed pad is an involution). Confidentiality: replace $\mathsf{AES\text{-}CTR}_{k_w}$ by a random function, at cost $\mathbf{Adv}^{\text{prf}}_{\mathsf{AES\text{-}CTR}}(\mathscr{B}')$; the resulting map is a length-preserving bijection applied to the AEAD ciphertext, and bijections cannot increase distinguishing advantage, so any distinguisher for the whitened scheme is a distinguisher for $\mathscr{E}$. Integrity: given a forgery $\widetilde{C}^*$ for the whitened scheme, $\mathscr{B}$ computes $C^* = \widetilde{C}^* \oplus \mathsf{AES\text{-}CTR}_{k_w}(\mathrm{IV}_j)$ — it can do so because $\mathscr{B}$ chooses k_w itself — and outputs C^*, which is a valid forgery for $\mathscr{E}$ and is fresh because $\mathscr{W}$ is injective. No PRF term appears in the authenticity bound because $\mathscr{B}$ simulates the whitening perfectly. □

Proposition 5 (Constant-time length leakage bound)**.** *With the padding rule (48), the wire length reveals only $\lceil (L+2)/256 \rceil$. Hence for plaintexts in a window of 256 bytes the observable length is identical, and the mutual information between L and the wire length obeys*

$$I\big(L;|\widetilde{C}|\big) \;\le\; \log_2 \left\lceil \tfrac{L_{\max}+2}{256} \right\rceil \; \textit{bits}, \tag{52}$$

independent of the plaintext distribution within each block class.

Proof. By (48), $|P'| = 256\lceil (L+2)/256 \rceil$ and $|\widetilde{C}| = |P'| + 16$, a deterministic function of the block index $B = \lceil (L+2)/256 \rceil$ alone. Therefore $|\widetilde{C}|$ is a deterministic function of B, and by the data processing inequality $I(L;|\widetilde{C}|) \le I(L;B) \le H(B) \le \log_2 B_{\max}$. □

All open-path comparisons use the constant-time primitive $\mathsf{ct_eq}(x,y) = \big(\bigvee_i (x_i \oplus y_i)\big) \stackrel{?}{=} 0$, evaluated without data-dependent branches or table lookups; padding is stripped only after tag verification succeeds.

5.3 The Counter-Free Forward Search Window

Definition 11 (Hyperbolic geodesic ratchet)**.** Fix an anchor $u_0 \in \mathbb{B}^n$ and a unit tangent $v_0 \in T_{u_0}\mathbb{B}^n$, and let

$$\gamma(t) = \exp_{u_0}(t\, v_0), \quad Q(\cdot) = \text{fixed-point quantizer, } 2^{-32} \text{ res.} \tag{53}$$

The chain key at step $j \ge 1$ is

$$K_j = \mathsf{HKDF\text{-}Expand}\Big(K_{j-1}, \text{“coros/ratchet”} \,\|\, Q\big(\gamma(j\delta)\big), 32\Big), \tag{54}$$

with step $\delta > 0$ and $K_0 = K$ from (45). No counter is transmitted on the wire.

Theorem 12 (Window correctness, cost and forgery bound)**.** *Let $\Omega = 32$. Algorithm 1 satisfies:*

Algorithm 1 COROS forward-window open (counter-free)

Input: envelope $\widetilde{C}$, associated data A, current index i, window $\Omega = 32$
State: K_i, geodesic (u_0, v_0, δ)
ok $\leftarrow 0$; $P^* \leftarrow \perp$; $j^* \leftarrow \perp$
for $j = i+1$ **to** $i+\Omega$ **do**
  $K_j \leftarrow \mathsf{HKDF\text{-}Expand}(K_{j-1}, \text{“coros/ratchet”} \,\|\, Q(\gamma(j\delta)), 32)$
  derive (k_a, k_w, n_0) from K_j by (47)
  $C \leftarrow \widetilde{C} \oplus \mathsf{AES\text{-}CTR}_{k_w}(\mathrm{IV}_j)$
  $r \leftarrow \mathsf{AES\text{-}256\text{-}GCM.Dec}(k_a, n_0 \oplus \mathrm{LE}_{96}(j), A, C)$
  $b \leftarrow \mathsf{ct_eq}(r \neq \perp, 1)$ {constant-time}
  $P^* \leftarrow \mathsf{ct_select}(b, r, P^*)$; $j^* \leftarrow \mathsf{ct_select}(b, j, j^*)$; ok $\leftarrow$ ok $\vee\, b$
end for
if ok $= 0$ **then return** $\perp$
commit K_{j^*}, set $i \leftarrow j^*$, strip padding via LE_{16} prefix
return P^*

(a) Correctness under loss. *If at most* $\Omega - 1$ *consecutive envelopes are lost or reordered ahead of the receiver, the next genuine envelope decrypts and the state resynchronizes to its true index.*

(b) Bounded cost. *Each open performs at most* Ω *KDF evaluations and* Ω *AEAD trial decryptions; the work is* $O(\Omega)$ *and, because the loop is executed to completion with constant-time selection, is independent of the successful index — no timing oracle on* j^* *exists.*

(c) Forgery bound. *For an adversary making* q *open queries against* AES-256-GCM *with* 128*-bit tags on messages of at most* ℓ *blocks,*

$$\mathbf{Adv}^{\text{forge}}_{\Omega\text{-}window} \;\le\; q\Omega \cdot \frac{\ell+1}{2^{128}} \;\approx\; q\Omega \cdot 2^{-128}. \tag{55}$$

For $\Omega = 32$ *this costs at most* $\log_2 \Omega = 5$ *bits of the authentication margin, leaving* ≥ 123 *bits.*

Proof. (a) The sender's index advances by one per envelope; if $g < \Omega$ envelopes were not received, the true index of the next genuine envelope is $i+g+1 \le i+\Omega$, which lies in the scanned range $\{i+1, \ldots, i+\Omega\}$. The chain (54) is deterministic given K_i and the geodesic, so the receiver reconstructs K_{i+g+1} exactly and AES-256-GCM.Dec succeeds. The commit step then sets $i \leftarrow i+g+1$. (b) The body executes exactly Ω times regardless of b — one HKDF-Expand, one AES-CTR expansion and one AEAD decryption each, all constant-time, with branch-free ct_select — so the work is $O(\Omega)$ and the trace is independent of j^*. (c) Each open query tests one forged envelope against Ω keys $k_a^{(j)}$ that are independent and uniform under the PRF assumption on HKDF-Expand; a fixed forgery verifies under a fixed AES-256-GCM key with 128-bit tag on ℓ blocks with probability $\le (\ell+1)2^{-128}$ [11]. A union bound over Ω trials and q queries gives (55). □

Remark 5 (Why counter-free)**.** Transmitting j would leak the message index to a passive observer and create a replay/skip primitive at the parser. The window makes the index an *inferred* quantity authenticated implicitly by AEAD success, at a provable cost of $\log_2 \Omega = 5$ bits and $O(\Omega)$ work. The geodesic term in (54) further binds each chain key to $\gamma(j\delta)$: a receiver whose hyperbolic anchor has diverged cannot derive valid keys even holding K_i, so phase-locking is enforced cryptographically.

6 Composite Certification

A deployment is *Atlas-conformant* iff the following eight gates hold simultaneously with the constants of Table 1: G_1: $K(\sigma) \in [-1/w_h, 0] \wedge \pi_1(\mathscr{M}) \cong \mathbb{Z}^n$ (Cor. 1); G_2: $\delta_{\max} < \pi \Rightarrow \widehat{W} = W$ (Lem. 2); G_3: W constant on connected perturbation sets (Thm. 3); G_4: $\kappa(T,t) \in (0,1)$ for all t (Thm. 5); G_5: $S_C S_R = s^2 \wedge S_C/S_R = \varphi^2$ (Thm. 6); G_6: $\sup_k L_k \leq e^{\rho} - 1 < e - 1$ (Lem. 3); G_7: $\mathbf{Adv}^{\text{ind-cca}} \leq \min\{\mathbf{Adv}_{\mathsf{ML\text{-}KEM\text{-}768}}, \mathbf{Adv}_{\mathsf{X25519}}\} + \mathbf{Adv}^{\text{prf}}$ (Thm. 10); G_8: $\mathbf{Adv}^{\text{forge}} \leq q\Omega 2^{-128}$ (Thm. 12). Each gate is an inequality over measurable quantities; no unverifiable assumption enters beyond MLWE hardness, Diffie–Hellman on Curve25519, and PRF security of HKDF-SHA-256. Calibration of $(T, w, s, \rho, \Lambda, \tau, \delta)$ is an empirical benchmarking exercise on a conformant implementation, not a soundness obligation.

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